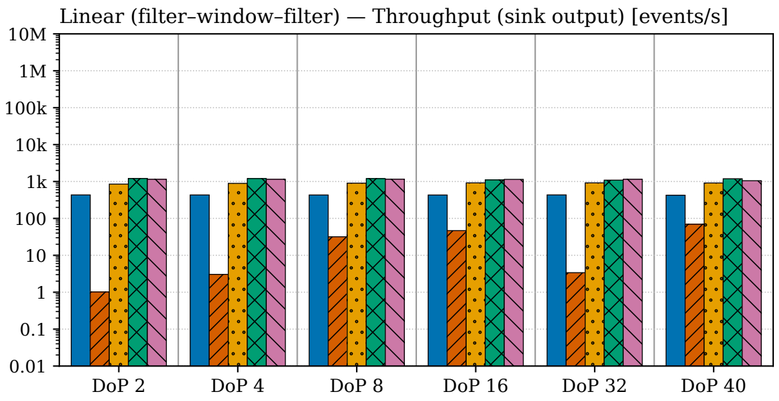
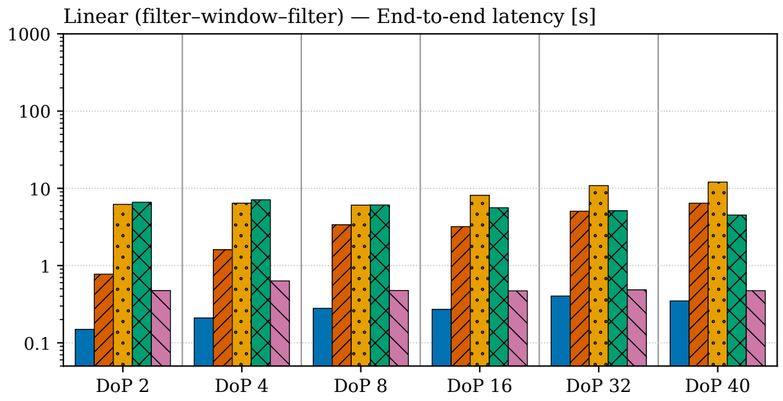
Event rate 500k events/s · at each DoP: one bar per system · × = no value

Flink · Storm · Spark · Kafka Streams · RisingWave

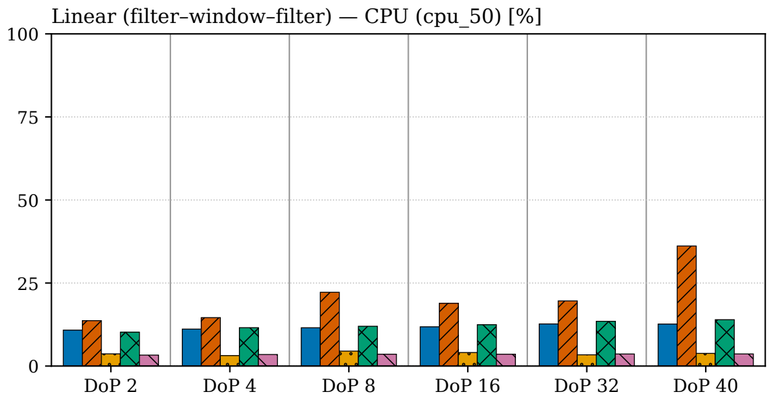
Linear (filter–window–filter) — Throughput (sink output) [events/s]



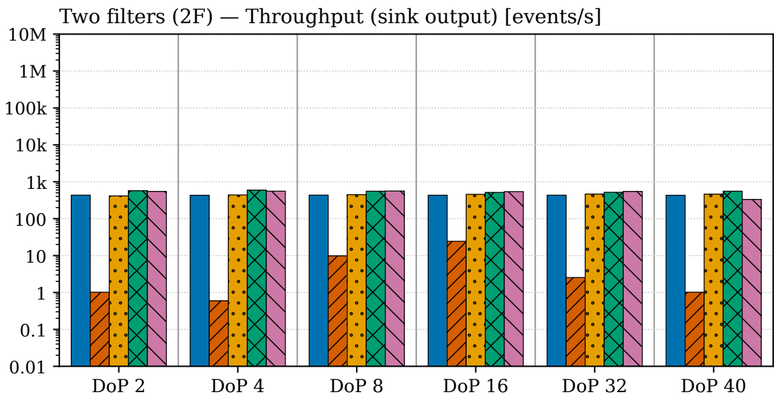
Linear (filter–window–filter) — End-to-end latency [s]



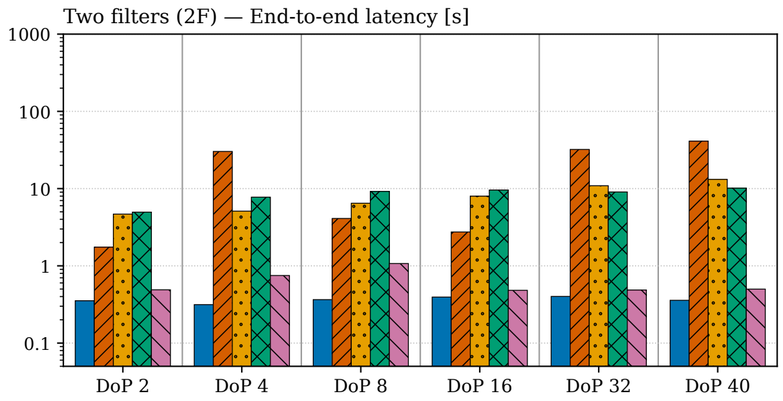
Linear (filter–window–filter) — CPU (cpu_50) [%]



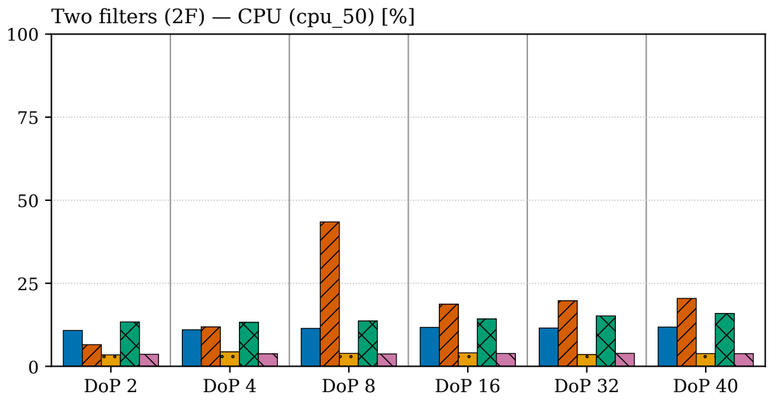
Two filters (2F) — Throughput (sink output) [events/s]



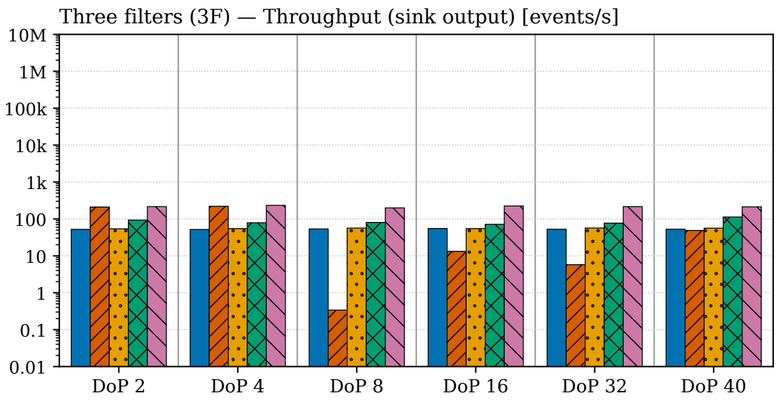
Two filters (2F) — End-to-end latency [s]



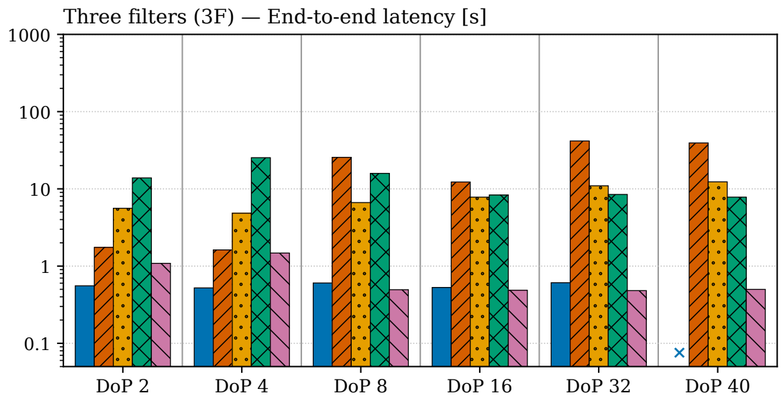
Two filters (2F) — CPU (cpu_50) [%]



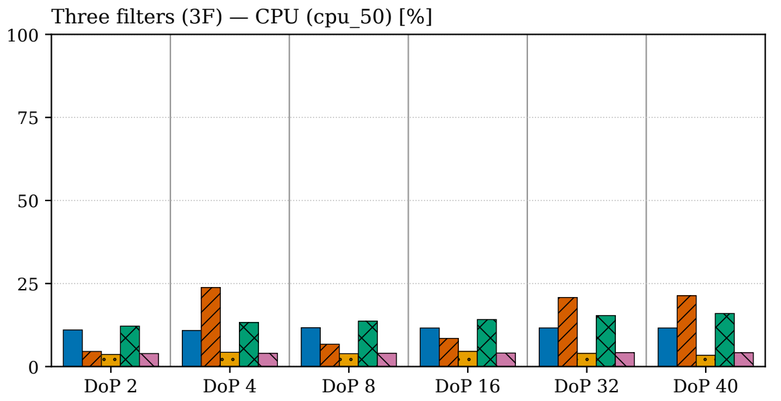
Three filters (3F) — Throughput (sink output) [events/s]



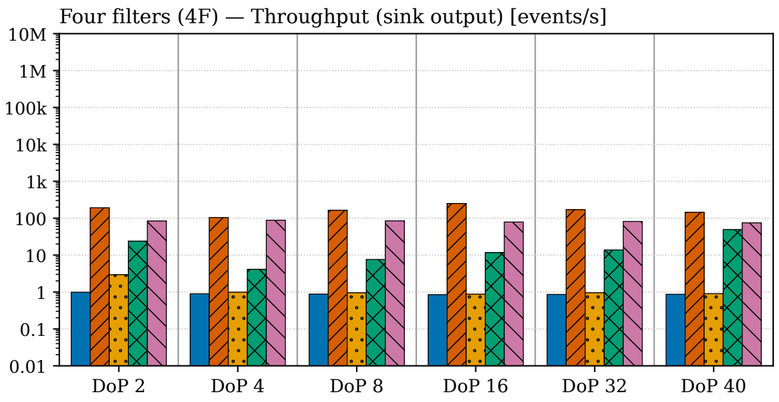
Three filters (3F) — End-to-end latency [s]



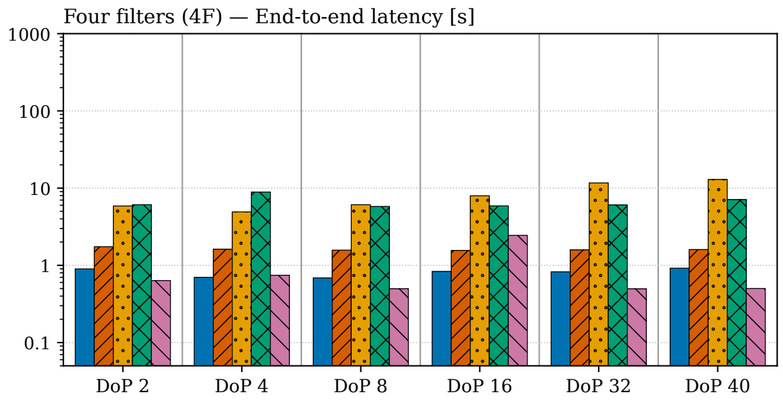
Three filters (3F) — CPU (cpu_50) [%]



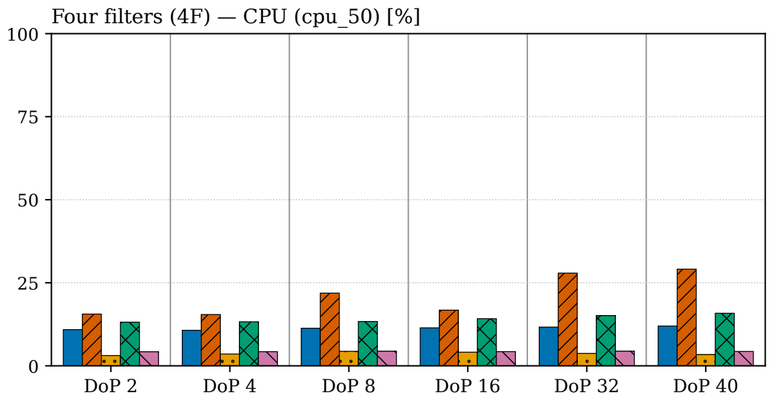
Four filters (4F) — Throughput (sink output) [events/s]



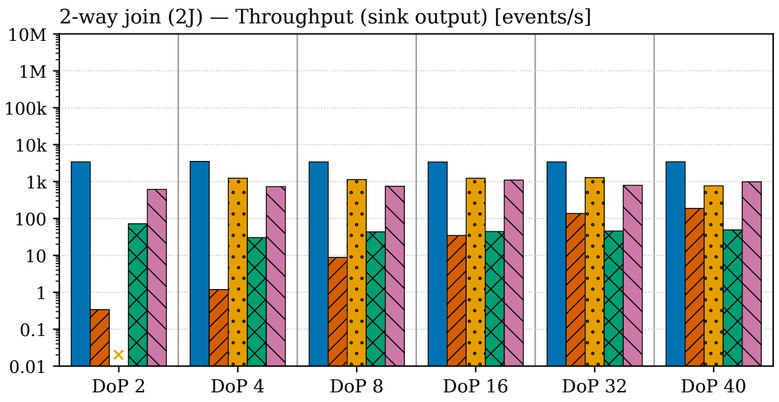
Four filters (4F) — End-to-end latency [s]



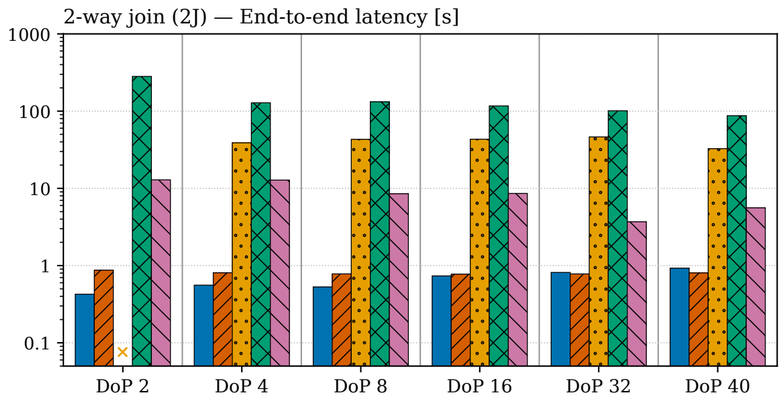
Four filters (4F) — CPU (cpu_50) [%]



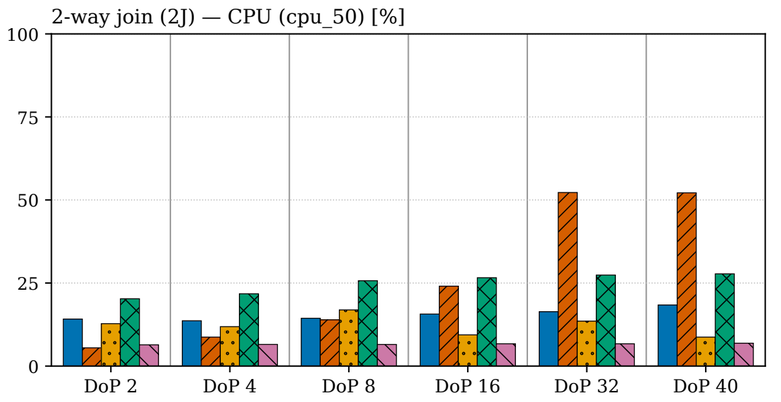
2-way join (2J) — Throughput (sink output) [events/s]



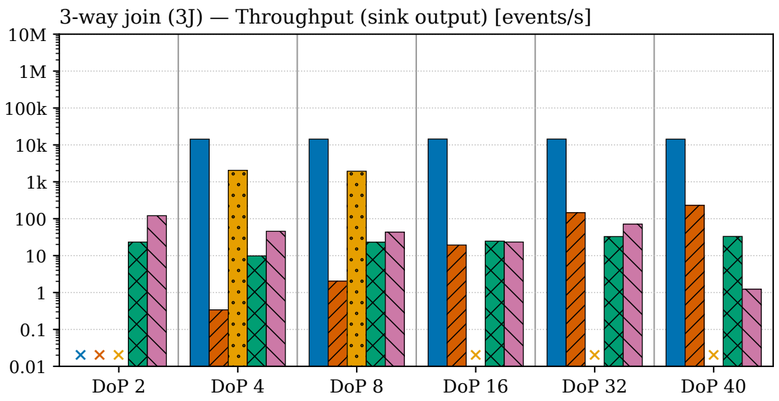
2-way join (2J) — End-to-end latency [s]



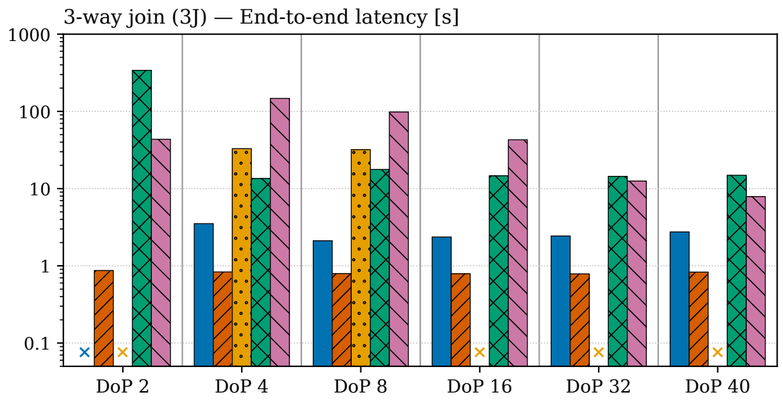
2-way join (2J) — CPU (cpu_50) [%]



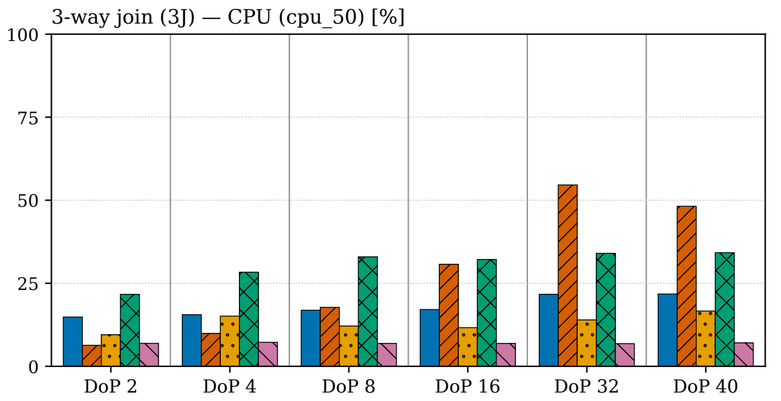
3-way join (3J) — Throughput (sink output) [events/s]



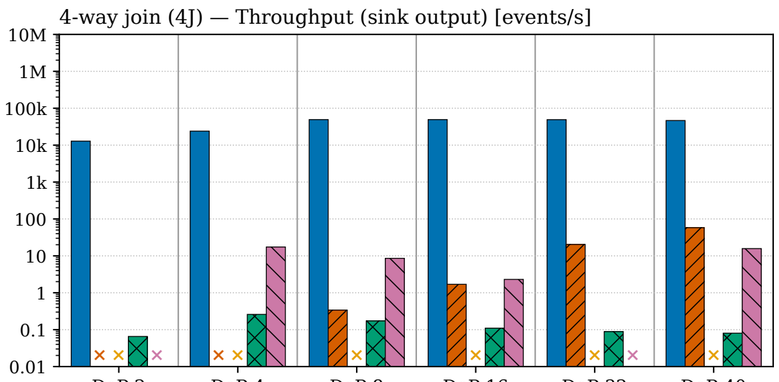
3-way join (3J) — End-to-end latency [s]



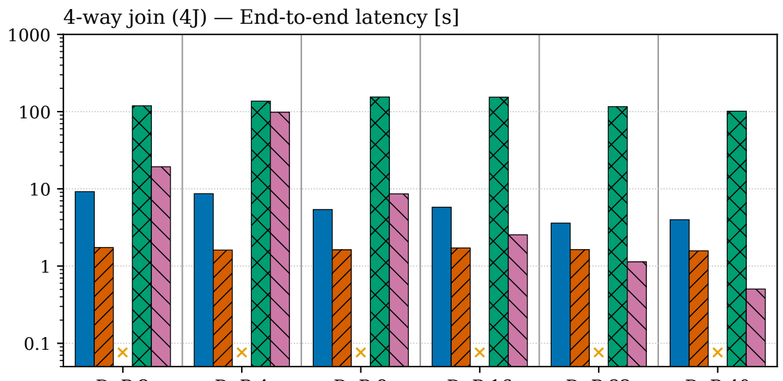
3-way join (3J) — CPU (cpu_50) [%]



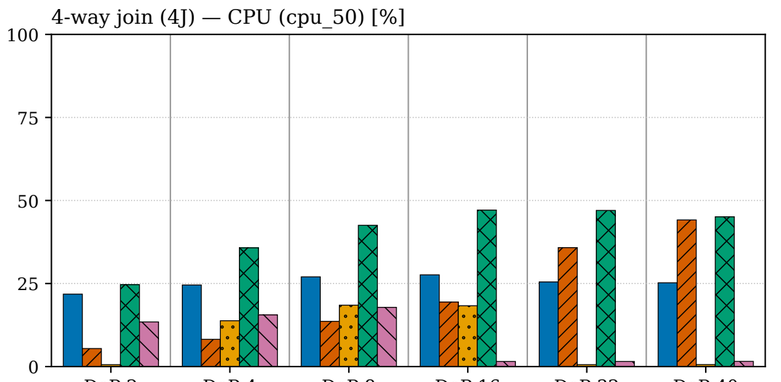
4-way join (4J) — Throughput (sink output) [events/s]



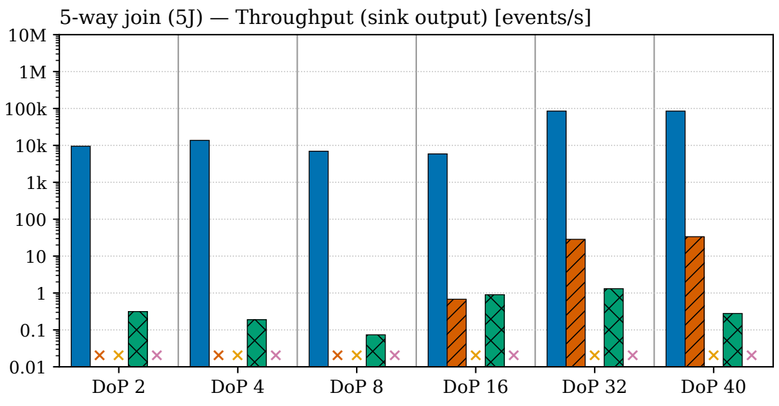
4-way join (4J) — End-to-end latency [s]



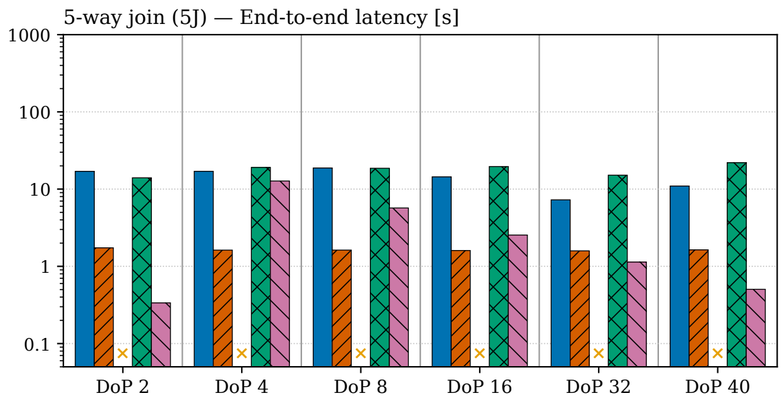
4-way join (4J) — CPU (cpu_50) [%]



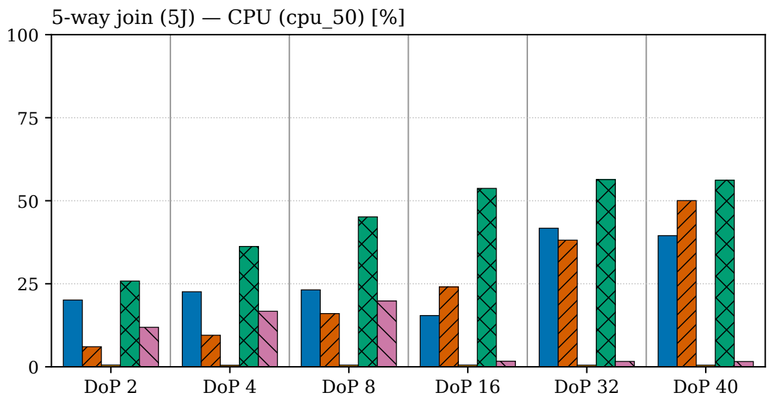
5-way join (5J) — Throughput (sink output) [events/s]



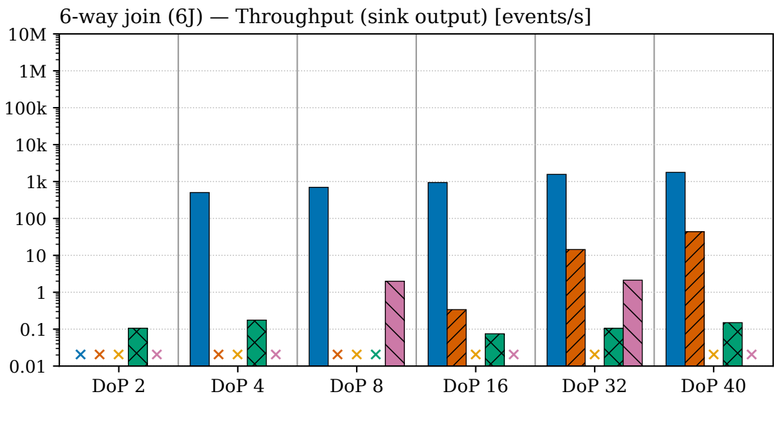
5-way join (5J) — End-to-end latency [s]



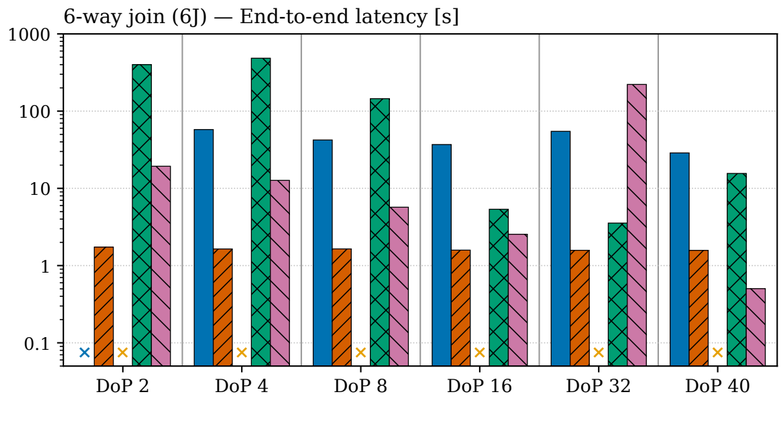
5-way join (5J) — CPU (cpu_50) [%]



6-way join (6J) — Throughput (sink output) [events/s]



6-way join (6J) — End-to-end latency [s]



6-way join (6J) — CPU (cpu_50) [%]

